Impulse and Step Responses in a Buck Converter with a Current PI Controller

Mini Paper — Power Electronics Fundamentals

Objective

This note analyzes the impulse and step responses of a buck converter operating under current PI control.
The focus is on:

- First-order inductor dynamics
- Second-order closed-loop behavior
- Relationship between PI gains and transient response

The goal is to connect classical system theory with practical current-loop behavior.

Time-Domain Response Framework

For a transfer function:

$$Y(s) = G(s)X(s)$$

Impulse input:

$$X(s) = 1$$

Step input:

$$X(s) = \frac{1}{s}$$

Impulse response reveals intrinsic system dynamics.
Step response reveals rise time, overshoot, damping, and steady-state behavior.

First-Order System

General form:

$$G(s) = \frac{K}{1 + sT}$$

Step response:

$$y(t) = K\left(1 - e^{-t/T}\right)$$

T = time constant

Settling time ≈ 4T

Monotonic response (no overshoot)

In a buck converter, inductor current dynamics are naturally first order.

Buck Converter Plant Model

Duty-to-current plant:

$$G_p(s) = \frac{1}{Ls + R}$$

If R is small:

$$G_p(s) \approx \frac{1}{Ls}$$

The inductor defines the dominant current dynamics.

PI Controller

$$G_c(s) = K_p + \frac{K_i}{s}$$

Open-Loop Transfer Function

$$G_{ol}(s) = G_c(s)G_p(s)$$

Substitute:

$$G_{ol}(s) = \left(K_p + \frac{K_i}{s}\right)\frac{1}{Ls}$$

Simplify:

$$G_{ol}(s) = \frac{K_p}{Ls} + \frac{K_i}{Ls^2}$$

Closed-Loop Transfer Function

For unity feedback:

$$G_{cl}(s) = \frac{G_{ol}(s)}{1+G_{ol}(s)}$$

Substitute and simplify:

$$G_{cl}(s) = \frac{K_p s + K_i}{Ls^2 + K_p s + K_i}$$

This is a standard second-order form.

Mapping to Standard Second-Order Form

Standard form:

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Compare denominator:

$$Ls^2 + K_p s + K_i$$

Divide by L:

$$s^2 + \frac{K_p}{L}s + \frac{K_i}{L}$$

Match coefficients:

$$\omega_n^2 = \frac{K_i}{L}$$

$$2\zeta\omega_n = \frac{K_p}{L}$$

Therefore:

$$\omega_n = \sqrt{\frac{K_i}{L}}$$

$$\zeta = \frac{K_p}{2\sqrt{LK_i}}$$

Interpretation

K_i sets loop speed through ω_n

K_p sets damping through ζ

Poor ratio of K_p to K_i leads to oscillation or sluggish response

Step Response Behavior

If:

$\zeta > 1$ → overdamped

$\zeta = 1$ → critically damped

$0 < \zeta < 1$ → overshoot

$\zeta = 0$ → oscillation

In current control:

Overshoot stresses MOSFETs

Low damping increases current ripple

Proper tuning ensures fast, stable tracking

Summary

• The inductor creates a first-order plant.
• PI control transforms the system into a second-order closed-loop system.
• K_i determines natural frequency.
• K_p determines damping ratio.
• Impulse and step responses provide direct insight into stability and tuning quality.

This links classical second-order theory directly to current-loop design in power converters.